Topics 9.1 – 9.7: MCQ Practice

1. Which of the following equations represents a reaction for which the standard entropy change is negative ($\Delta S° < 0$) ?

   (A) $2\ H_2(g) + O_2(g) \rightarrow 2\ H_2O(g)$

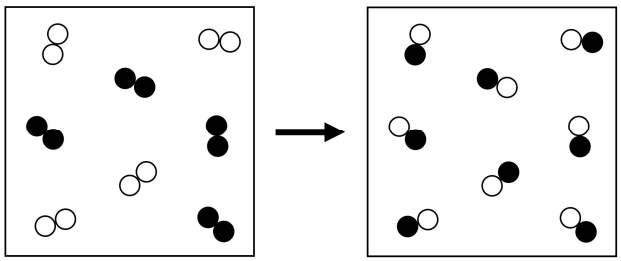
   (B) $2\ C_2H_6(g) + 7\ O_2(g) \rightarrow 4\ CO_2(g) + 6\ H_2O(g)$

   (C) $Mg(s) + 2\ HCl(aq) \rightarrow MgCl_2(aq) + H_2(g)$

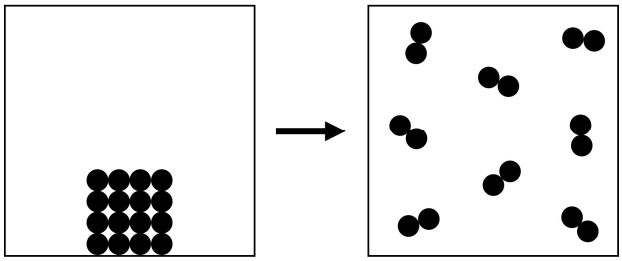
   (D) $(NH_4)_2CO_3(s) \rightarrow 2\ NH_3(g) + H_2O(g) + CO_2(g)$

2. Which of the following particulate representations represents the change with the greatest positive value of $\Delta S°$?

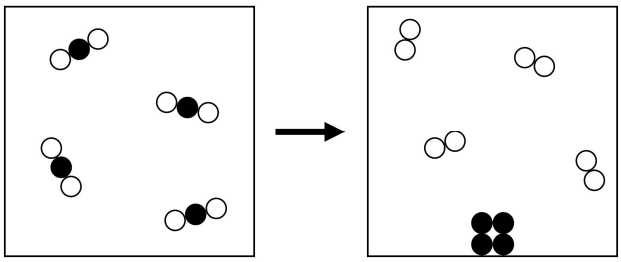
(A)



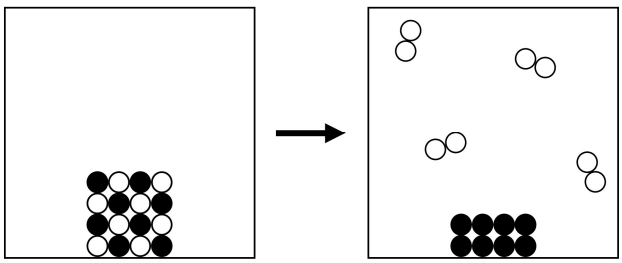
(B)



(C)



(D)



3. The values of the standard molar entropies of three different substances are listed in the following table.

Substance	Standard Molar Entropy (J/(K·mol))
$Cl_2(g)$	223
$NO(g)$	211
$NOCl(g)$	262

Based on this information, which of the following is closest to the value of $\Delta S°$ for the reaction represented by the following equation?

$$Cl_2(g) + 2\ NO(g) \rightarrow 2\ NOCl(g) \qquad \Delta S° = ?$$

(A) $-172\ J/(K \cdot mol_{rxn})$

(B) $-121\ J/(K \cdot mol_{rxn})$

(C) $+121\ J/(K \cdot mol_{rxn})$

(D) $+172\ J/(K \cdot mol_{rxn})$

4. The formation of gaseous hydrogen bromide from hydrogen gas and bromine vapor is represented by the following equation.

$$H_2(g) + Br_2(g) \rightarrow 2\ HBr(g) \qquad \Delta S° = 20\ J/(K \cdot mol_{rxn})$$

The values of the standard molar entropies of the reactants are listed in the following table.

Substance	Absolute Entropy (J/(K·mol))
$H_2(g)$	130
$Br_2(g)$	250

Based on this information, which of the following is closest to the value of the standard molar entropy of $HBr(g)$?

(A) 180 J/(K·mol)

(B) 200 J/(K·mol)

(C) 380 J/(K·mol)

(D) 400 J/(K·mol)

$$CO(g) + O_3(g) \rightarrow CO_2(g) + O_2(g) \quad \Delta H° = -426 \text{ kJ/mol}_{rxn}; \quad \Delta S° = -18 \text{ J/(K·mol}_{rxn})$$

5. The reaction between $CO(g)$ and $O_3(g)$ is represented by the equation shown above. Which of the following is most likely to be true about this reaction at 298 K?

   (A) $\Delta G° < 0$ and $T\Delta S°$ is smaller in magnitude than $\Delta H°$.

   (B) $\Delta G° < 0$ and $T\Delta S°$ is larger in magnitude than $\Delta H°$.

   (C) $\Delta G° > 0$ and $T\Delta S°$ is smaller in magnitude than $\Delta H°$.

   (D) $\Delta G° > 0$ and $T\Delta S°$ is larger in magnitude than $\Delta H°$.

6. Which of the following reactions is not thermodynamically favored at low temperatures but becomes favored as the temperature increases?

	Reaction	$\Delta H°$ kJ/mol$_{rxn}$	$\Delta S°$ J/(K·mol$_{rxn}$)
(A)	$2\ CO(g) + O_2(g) \rightarrow 2\ CO_2(g)$	–566	–173
(B)	$2\ H_2O(g) \rightarrow 2\ H_2(g) + O_2(g)$	484	90.0
(C)	$2\ N_2O(g) \rightarrow 2\ N_2(g) + O_2(g)$	–164	149
(D)	$PbCl_2(s) \rightarrow Pb^{2+}(aq) + 2\ Cl^-(aq)$	23.4	–12.5

7. When water is added to a mixture of $Na_2O_2(s)$ and $S(s)$, a redox reaction occurs, as represented by the equation below.

   $2\ Na_2O_2(s) + S(s) + 2\ H_2O(l) \rightarrow 4\ NaOH(aq) + SO_2(aq) \quad \Delta H° = -610 \text{ kJ/mol}_{rxn}; \quad \Delta S° = -7.3 \text{ J/(K·mol}_{rxn})$

   Which of the following statements about the thermodynamic favorability of the reaction at 298 K is correct?

   (A) It is thermodynamically unfavorable.

   (B) It is thermodynamically favorable and is driven by $\Delta S°$ only.

   (C) It is thermodynamically favorable and is driven by $\Delta H°$ only.

   (D) It is thermodynamically favorable and is driven by both $\Delta H°$ and $\Delta S°$.

$$CO(g) + 2\,H_2(g) \rightarrow CH_3OH(g) \qquad \Delta H^\circ < 0 \text{ and } \Delta G^\circ < 0 \text{ at } 298\text{ K}$$

8. A 1.0 mol sample of $CO(g)$ and a 1.0 mol sample of $H_2(g)$ are placed in a previously evacuated 10.0 L vessel at 298 K. Even though the reaction represented by the equation above is thermodynamically favorable, the formation of $CH_3OH(g)$ is not observed. Which of the following best explains why $CH_3OH(g)$ is not produced at 298 K?

   (A) At 298 K, the energy of most molecular collisions is lower than the activation energy for the reaction.

   (B) At 298 K, some $CH_3OH(g)$ must first be present in the vessel in order for the reaction to occur.

   (C) The $CO(g)$ and the $H_2(g)$ must be mixed in a 1-to-2 mole ratio in order for the reaction to occur.

   (D) The rate of the reaction is extremely slow because the value of ΔH° is negative.

$$2\,H_2(g) + O_2(g) \rightarrow 2\,H_2O(g) \qquad \Delta H^\circ = -484\text{ kJ/mol}_{rxn}\,; \quad \Delta S^\circ = -89\text{ J/(K}\cdot\text{mol}_{rxn}\text{)}$$

9. When $H_2(g)$ and $O_2(g)$ are mixed together in a rigid reaction vessel at 298 K, no reaction occurs. When the mixture is sparked, however, the gases react vigorously according to the equation above. Which of the following statements correctly explains why the spark is needed for the reaction to occur when the gases are originally at 298 K?

   (A) The reaction is not thermodynamically favorable at 298 K.

   (B) The reaction rate is very slow at 298 K because $\Delta H^\circ < 0$.

   (C) The reaction rate is very slow at 298 K because $\Delta S^\circ < 0$.

   (D) The reaction has a large activation energy at 298 K.

$$FeF_2(s) \rightleftarrows Fe^{2+}(aq) + 2\,F^-(aq) \qquad K_1 = 2 \times 10^{-6}$$

$$F^-(aq) + H^+(aq) \rightleftarrows HF(aq) \qquad K_2 = 1 \times 10^{3}$$

$$FeF_2(s) + 2\,H^+(aq) \rightleftarrows Fe^{2+}(aq) + 2\,HF(aq) \qquad K_3 = \ ?$$

10. On the basis of the information above, the dissolution of $FeF_2(s)$ in acidic solution is

    (A) thermodynamically favorable, because $K_2 > 1$

    (B) thermodynamically favorable, because $K_3 > 1$

    (C) <u>not</u> thermodynamically favorable, because $K_1 < 1$

    (D) <u>not</u> thermodynamically favorable, because $K_3 < 1$

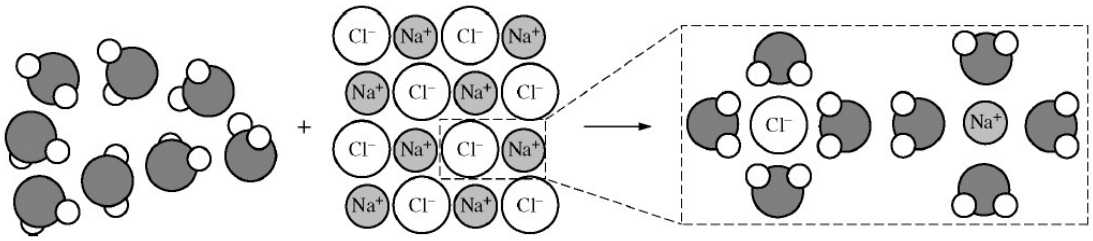
11. A reaction is known to have a negative value of $\Delta H°$ and a negative value of $\Delta S°$. Which of the following statements correctly describes the equilibrium constant, K_{eq}, for this reaction?

    (A) K_{eq} will be less than 1 at all temperatures.

    (B) K_{eq} will be greater than 1 at all temperatures.

    (C) K_{eq} will be greater than 1 only at low temperatures.

    (D) K_{eq} will be greater than 1 only at high temperatures.



12. The process of dissolution of NaCl(s) in $H_2O(l)$ is represented in the diagram above. Which of the following summarizes the signs of $\Delta H°$ and $\Delta S°$ for each part of the dissolution process?

	Breaking solvent–solvent interactions		Breaking solute–solute interactions		Forming solute–solvent interactions	
	$\Delta H°$	$\Delta S°$	$\Delta H°$	$\Delta S°$	$\Delta H°$	$\Delta S°$
(A)	+	+	+	+	–	–
(B)	+	+	+	+	–	+
(C)	–	–	–	–	+	+
(D)	–	+	–	+	+	–

Reaction 1: $4\ NH_3(g) + 5\ O_2(g) \rightarrow 4\ NO(g) + 6\ H_2O(l)$ $\Delta G° = -1011\ kJ/mol_{rxn}$

Reaction 2: $2\ NO_2(g) \rightarrow 2\ NO(g) + O_2(g)$ $\Delta G° = 71\ kJ/mol_{rxn}$

Reaction 3: $4\ NO_2(g) + O_2(g) + 2\ H_2O(l) \rightarrow 4\ HNO_3(aq)$ $\Delta G° = -166\ kJ/mol_{rxn}$

13. Based on the values of $\Delta G°$ for the three reactions represented above, what is the value of $\Delta G°$ for the reaction represented below?

$$4\ NH_3(g) + 8\ O_2(g) \rightarrow 4\ HNO_3(aq) + 4\ H_2O(l)$$

(A) $-1035\ kJ/mol_{rxn}$

(B) $-1106\ kJ/mol_{rxn}$

(C) $-1248\ kJ/mol_{rxn}$

(D) $-1319\ kJ/mol_{rxn}$